

Spotted Python

Week 2 Tutorial

- ★ Markdown
- ★ Modules
- ★ Lists and Arrays

Markdown

- ★ Text-to-HTML conversion tool
- ★ Can be used in Jupyter Notebooks for annotation of code

Element	Markdown Syntax
Heading	# Heading 1
Bold	**Bold text** or __Bold text__
Italic	<i>*Italic text*</i> or <i>_Italic text_</i>
Highlight	==Highlighted text==
Unordered List	- First item
Link	[title](https://www.example.com)

- ★ Allows mathematical expressions in LaTeX-style syntax
- ★ Cheat sheet: <https://www.markdownguide.org/cheat-sheet/>
- ★ Complete **Exercise 1**

BODMAS with Python

★ Operators in Python are applied much like the rules we learnt in primary school. Specifically:

1. () Brackets
2. ** (not ^) Exponentiation
3. *, /, //, % Multiplication, Division, Floor Division, Modulus
4. +, - Addition, Subtraction

★ Complete **Exercise 2**

Note:

Using comments and easily interpreted variable names makes code easier to read. But make sure your variable name starts with a letter and is not a reserved word within Python.

Modules

- ★ A file containing Python definitions and statements that can be imported into another script
- ★ Common modules and how they are imported:
 - ★ `import numpy as np`
 - ★ Manipulation of arrays
 - ★ `import matplotlib.pyplot as plt`
 - ★ Create static, animated and interactive visualisations
 - ★ `from astropy.io import fits`
 - ★ Contains range of tools for astronomical computations
 - ★ FITS (Flexible Image Transport System) is a digital file format containing multi-dimensional arrays

Data Types

- ★ A file of different data types:

Data Type	Meaning
Integer	No decimal point
Floating point	With decimal
Complex number	
String	Text

- ★ Cannot add or subtract strings with an integer
- ★ To work with the data types, you can convert them into the same type

Lists and Arrays

- ★ Lists are a sequence of objects enclosed in `[]` and separated by commas
- ★ Arrays are an arrangement of numbers or objects formatted into rows and columns
- ★ Indexing:
- ★ Complete **Exercise 3**

	1	3	5	7	9
index	0	1	2	3	4
negative index	-5	-4	-3	-2	-1

Notice that indexing starts from 0

	0	1	2	
0	(0,0)	(0,1)	(0,2)	← Column Index
1	(1,0)	(1,1)	(1,2)	
2	(2,0)	(2,1)	(2,2)	
← Row Index				